

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

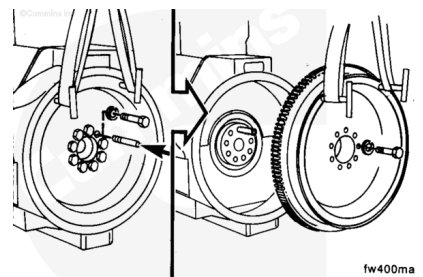
- Disconnect the batteries. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Disconnect the starter air to prevent accidental engine starting. Refer to Procedure 012-022 (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>).
- Remove the transmission, clutch, and all related components. Refer to the OEM manual.



Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



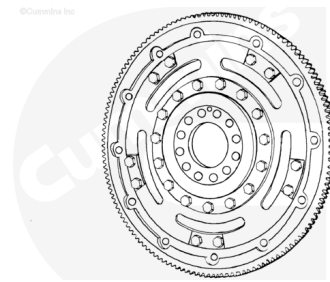
Note : The use of a guide pin will aid the disassembly procedure.

Remove the SAE Grade 8 capscrews and the hardened washers. Remove the flexplate.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the risk of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

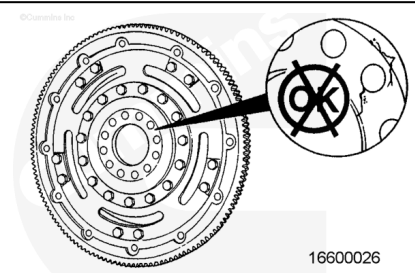
Do **not** disassemble the flexplate assembly. Refer to Procedure 016-999 (</qs3/pubsys2/xml/en/procedures/28/28-016-999.html>).

Use solvent. Clean the flexplate. Dry with compressed air.

Check the ring gear for damaged or worn teeth. If the ring gear is damaged or worn, it **must** be replaced. Refer to Procedure 016-008 (</qs3/pubsys2/xml/en/procedures/28/28-016-008.html>).

⚠ WARNING ⚠

Cummins Inc, does not recommend resurfacing of the flexplate. Resurfacing of the flexplate can cause possible personal injury, property damage, or cause it to be too thin and result in breakage.



Use the dye penetrant method to check the area of the mounting holes for cracks. There **must not** be any cracks in the flexplate mounting area.

Check the mounting pilot for damage. If the pilot is damaged, the flexplate **must** be replaced.

Check the tightness of the capscrews.

For standard flexplate 25 capscrews:

Torque Value: 290 n•m [215 ft-lb]

Use SAE Grade 8 capscrews if replacement is required.

For insulated flexplate:

Torque Value:

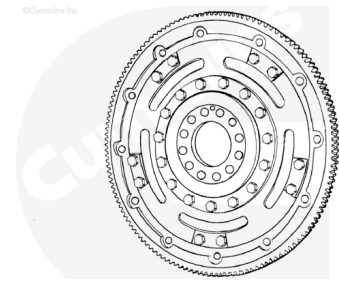
4 outer flexplate capscrews 293 n•m [216 ft-lb]

Torque Value:

15 inner flexplate capscrews 293 n•m [216 ft-lb]

If replacement is required, use **only** the correct grade 10.9 capscrews released by Cummins®.

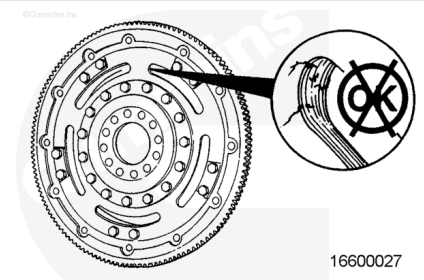
If the capscrews require replacement, remove and replace them one at a time. Do **not** disassemble the flexplate.



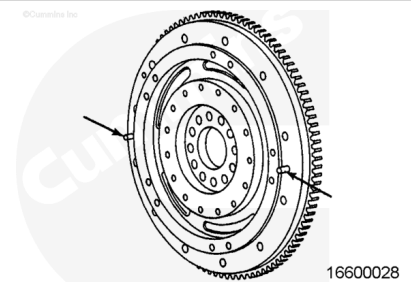
The flexplate can consist of as many as nine separate steel plates.

Check closely for cracks.

If any of the steel plates are cracked or damaged, the engine flexplate **must** be replaced.



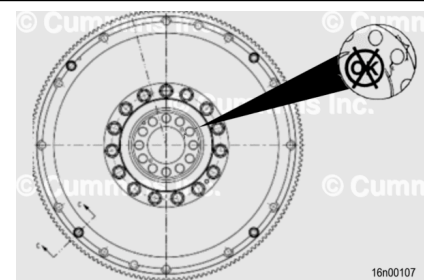
Check the two tapered dowels for damage. If the dowels are bent or broken, they **must** be replaced.



For engines using insulated flexplate, use the dye penetrant method to check the area of the mounting holes for cracks. There **must not** be any cracks in the flexplate mounting area.

Mechanical damage can be caused to the flexplate because of impact from heavy subjects like tools, or due to overloading. If any of the following are present, replace the insulated flexplate immediately:

- Visible fiber failures or delamination (layer separation).
- Audible cracking during slow rotation of the part.
- Any discoloration of the resin connecting the central hub to the carbon fiber section.



Measure

Note : The crankshaft must be pushed or pulled in the same direction each time a measurement is made, in order to obtain an accurate reading.

Bar the engine from the front of the crankshaft for both radial and face measurements. The barring device will cause the flexplate to distort if barred from the rear, and also cause the crankshaft to walk up the main bearing sidewall.

Cummins recommends to install a flywheel whenever inspecting the flywheel face and outer runouts. In situations that do **not** allow for installation of a flywheel and the **only** method of gaining flywheel face and outer runouts is via the flexplate option. Barring the engine from the rear of the engine via the flexplate may damage the flexplate and crankshaft main bearings. The engine **must** be barred from the front of the engine via a crankshaft mounted adaptor suitably designed for the application, space constraints and local health and safety protocols in which the inspection takes place.

Radial Runout

Attach an indicator to the flywheel housing with the indicator tip on the flexplate as shown. Check the indicator while rotating the engine.

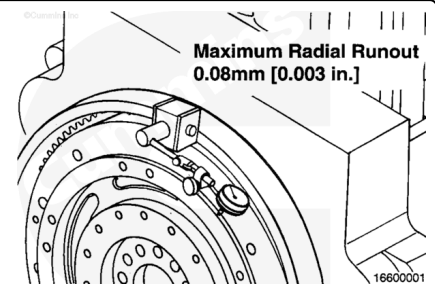
For standard flexplate:

Maximum Radial Runout: 0.08 mm [0.003 in]

For insulated flexplate:

Maximum Radial Runout: 0.25 mm [0.010 in]

If the runout is **not** within specification, the pilot on the flexplate is **not** seated correctly on the crankshaft. If the pilot is damaged, the entire flexplate assembly **must** be replaced.



Face Runout

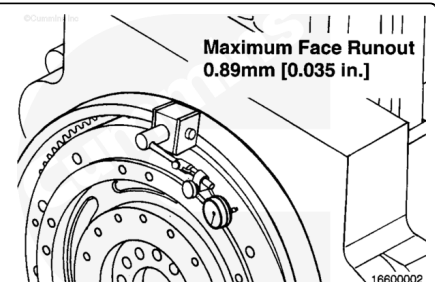
Attach an indicator to the flywheel housing with the indicator tip on the face of the face of the flexplate. Check the indicator while rotating the engine.

For standard flexplate:

Maximum Face Runout: 0.89 mm [0.035 in]

For insulated flexplate:

Maximum Face Runout: 1.5 mm [0.06 in]



If the alignment is **not** within specification, check for foreign material between the flexplate and the crankshaft.

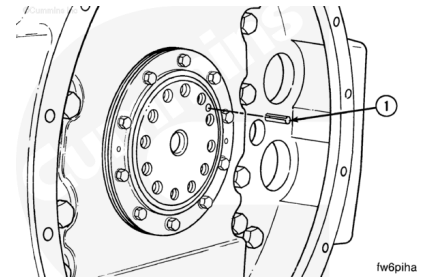
Two 3/8 x 16 inch threaded holes were added to the adapter to aid in the removal of the assembly.

Install

Install the roll pin (1) in the crankshaft.

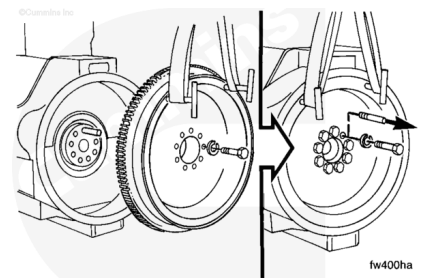
The pin is required to align the timing marks that are on the flexplate.

Note : The flexplate is not the same for the K38 and K50 engines. The valve and injector adjustment marks that are stamped on the parts are different.



⚠ WARNING ⚠

The flexplate mounting capscrews must be SAE Grade 8 minimum with rolled threads. The flexplate mounting plain washers are special hardened . Do not substitute the parts or a failure can result that can cause personal injury.



The use of a guide bolt will aid the assembly procedure.

Do **not** lubricate the threads of the crankshaft.

Use engine oil. Lubricate the capscrews and the washers. Allow the excess oil to drip from the parts.

Check to be sure the mating surfaces of the crankshaft and flexplate are clean, dry, and free of nicks or burrs.

Align the hole in the flexplate for the roll pin with the roll pin in the crankshaft.

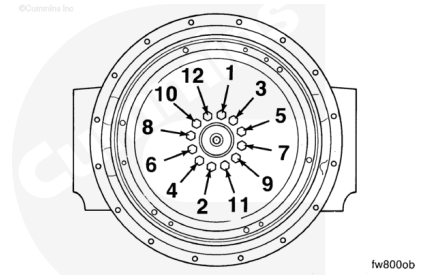
Install the flexplate, washers, and the capscrews. The flexplate **must** fit firmly against the crankshaft.

Use the following steps to tighten the capscrews in the sequence shown.

Torque Value:

1. 150 n•m [110 ft-lb]
2. 285 n•m [210 ft-lb]

3. 515 n•m [380 ft-lb]



Finishing Steps

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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Connect the starter air. Refer to Procedure 012-022 (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>).
- Start the engine. Check equipment operation.